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NEPAL COLLEGE OF INFORMATION TECHNOLOGY
ASSESSMENT FALL- 2025

Level: Bachelor
Programme: BE CE
Course: Probability and Statistics
3rd Semester

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3 hours

Candidates are required to give their answer in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempts all the questions:

1. a) An investigator wants to study the speed of cars and collected the speed of 30 vehicles

35	37	42	45	47	48	50	55	67	70
75	80	90	95	94	48	55	60	71	63
70	65	80	55	40	35	36	85	79	30

i) present the above data in stem and leaf display
ii) construct box and whisker plot
iii) construct the continuous class interval
iv) Find median speeds; find 25% and 75% speed of the vehicles.

b) A purchasing agent obtained samples of bulbs from two suppliers .He had the samples tested in his own laboratory for length of life with the following results

Length of life (hrs)	Sample from A	Sample from B
700-900	10	3
900-1100	16	42
1100-1300	26	12
1300-1500	8	3

a) Which company's bulb gives a higher average life?
b) Which company's bulbs are more uniform?

2. a) State Baye's theorem? In a class of 75 students, 15 students were considered to be intelligent, 45 as medium and rest below the average. The probability that a very intelligent students fails in examination is 0.005, the medium student failing has probability 0.05 and below the average is 0.15. If a student is known to have passed the examination, what is the probability that he is below the average?

b) An urn contains 7 red and 4 white balls, 3 balls are drawn at random, find the probability distribution of number of white balls drawn. Also find the mathematical expectation and variance of the distribution.

3. a) The number of accidents in a year attributed to taxi driver in a city is Poisson distribution with mean is 3. Out of 1000 taxi driver, find approximately the number of driver with
   i. more than 3 accidents in a year
   ii. atleast 1 accident

b) Suppose that the joint probability distribution of X and Y is given as

$f(x,y) = 4xy;\ 0<x<1,\ 0<y<1$

;0 otherwise

i. verify the probability density function
ii. find marginal probability function of X and Y
iii. find conditional probability function of X and Y
iv. Are X and Y are independent.

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4. a) In a statistics examination 15% of the candidates scored more than 63 marks and 10% of the candidates scored less than 32 marks. Assuming that the marks were distributed normally find the mean and standard deviation? Determine the $P(20<X<80)$

b) A random sample of 12 values from a normal population showed a mean of 31.9 inches and sum of square of deviation from this mean is 125 inches. Obtain 95% and 99% confidence interval for population mean?

5. a) Memory capacity of 10 students was tested before and after training, state whether the training was effective or not at 5% level of significance.

Roll no:	1	2	3	4	5	6	7	8	9	10
Before training:	12	14	11	8	7	10	3	0	5	6
After training:	15	16	10	7	5	12	10	2	3	8

b) The coordinator in a college claimed that atleast 98% of the students submit their assignment on time. Taking the sample of 250 students, 15 were not submitting assignment in whole semester. Test his claim at 10% level of significance

6. a) The following data related to the number of children classified according to the type of feed and the nature of teeth.

Type of feed	Nature of teeth	
	normal	Defective
Breast	18	12
bottle	2	13

Do the information provide sufficient evidence to conclude that type of feeding and nature of teeth are dependence? Use chi square test at 5% level of significance

b) A panel of two Judges A and B graded 7 dramatic performance by independently awarding marks as follows

Marks A (x)	46	42	44	40	43	41	45
Marks B (y)	40	38	36	35	39	37	41

i. Fit a straight line on $y=a+bx$
ii. Estimate the value of y when $x=37$
iii. Find standard error of estimate
iv. Find the coefficient of determination

7. Write short notes on; (any two)
   a) Point and interval estimation
   b) Condition to apply binomial and poisson distribution
   c) Characteristics of Beta and gamma distribution

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POKHARA ENGINEERING COLLEGE

Level: Bachelor
Programme: BE Computer/IT
Course: Probability and Statistics (New)

Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) From the following frequency distribution, 7

Obtained Marks	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students	7	10	17	20	8	6	2

Construct an ogive that will help you the answer to find the number of students securing:

i. Less than 35 marks ii. Between 25 and 60 marks
iii. More than 15 marks

b) The life of two models A and B of Polythene bags in a recent survey is shown below. 8

Bursting pressure (in lb.)	No. of Polythene bags	
	A	B
5-10	2	9
10-15	9	11
15-20	29	18
20-25	44	32
25-30	11	17
30-35	5	13

i. What is the average life of each model?
ii. Which model has greater uniformity and why?

2. a) A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is 1/7 and that of wife's selection is 1/5. What is the probability that 7

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i. both of them will be selected. ii. only one of them will be selected.
iii. none of them will be selected. iv. husband will be selected but not wife.

b) In a college 45% student belong to civil, 30% belong to electronics and remaining to other faculties. The probability of being top is 5%, 4% and 2% respectively in civil, electronics and others. If this year's result is published, what is the probability that the topper is from electronics? 8

3. a) A random variable X has the following probability function. 8

X	2	3	4	5	6	7
F(x)	0.1	k	0.2	2k	0.3	k

Find the value of k and calculate mean, S.D., $E(2x - 9)$ and $V(5x + 2)$.

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b) An office switchboard receives telephone calls at the rate of 3 calls per minute on an average. If receiving of calls follows a Poisson distribution, find the probability of receiving: 7

i. no calls in one minute interval. ii. at least 3 call in a one minute interval.
iii. at most 2 calls in 5 minute interval.

4. a) Of a large group of men, 5% are under 60 inches in height and 40% are between 60 and 65 inches. Assuming normal distribution, find the mean and standard deviation. 7

b) Joint probability distribution of X and Y is given by; 8

$f(x,y) = 4xye^{-(x^2+y^2)}; x > 0, y > 0$

i. Test whether X and Y are independent or not.
ii. Find the conditional density of X given Y=y.

5. a) The quality control Engineer at a light bulb factory needs to estimate the average life of a large shipment of light bulbs. The process standard deviation is known to be 100 hours. A random sample of 25 light bulbs indicated a sample average life of 350 hours. 7

i. Calculate the standard error of the mean.
ii. Set of 95% confidence limits for the true average life of light bulbs in this shipment.

b) Two research laboratories have independently produced drugs that provide relief of cough. The first drug was tested on a group of 90 cough sufferers and produced an average of 8.5 hrs relief and a sample standard deviation of 1.8 hrs. The second drug was tested on 80 cough sufferers, producing an average of 7.9 hrs of relief and a sample standard deviation of 2.1 hrs. At a 0.05 level of significance, does the second drug provide a significantly shorter period of relief? 8

6. a) An I.Q. test was administered to 5 persons before and after they were trained. The result are given below: 7

Candidates	1	2	3	4	5
IQ before Training	110	120	123	132	125
IQ after Training	120	118	125	136	121

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Is the training effective at 5% level of significance?

b) The following table gives the age (X) and blood pressure (Y) of 7 women. 8

X	56	47	42	60	72	63	55
Y	147	128	140	155	160	149	145

i. Find the correlation coefficient between age and blood pressure and interpret it.
ii. Find the regression equation of Y on X and interpret the slope coefficient.

7. Write short notes on: **(Any two)** 2x5

a) Sources of data
b) Probability mass function and probability density function
c) Properties of good estimator

The End

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Lumbini Engineering, Management & Science College
Final Internal Exam 2026

Level: Bachelor **Time: 3 hrs**

Program: BE Computer 5th Sem. **Full Mark:100**

Course: Probability & Statics **Pass Mark:45**

1. From the pair of shares x & y below, state which is more consistent. (8)

x: 55	54	53	56	58	52	50	51	49	52
y: 108	107	105	105	106	107	104	103	104	101

2. State Baye's theorem. In a bolt factory machines A, B & C manufacture respectively 25%, 30% & 40% of the total of their output 5%, 4%, 2% are defective bolts. A bolt is drawn at random from the product & found to be defective. What is the probability that it was manufactured by machine B. (7)

3. The mean & variance of binomial distribution are 3 & 2 respectively. Find the probability that the variable takes values (a) less than or equals to 2 (b) greater than or equal to 7. (7)

4. In a normal distribution 31% of the items are under 45 & 8% are over 64. Find the mean & s.d. of the distribution. (8)

5. Two random variable x & y have the following probability density function. (8)

$$F(x,y)=\begin{cases} k\,(4-x-y) & 0<x,2, \quad 0<y<2 \\ 0\ otherwise \end{cases}$$

Find (a) The value of k.

(b) Marginal density function of x & y.

(c) Conditional Probability density function of x given y & of y given x.

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6. Define Beta distribution. In a certain hills, the proportion of soil lapsed in 10 years in a random variable having beta distribution with $\alpha=3$ & $\beta=2$. (7)

(i) Find on average what percent of soil lapse in a given 10 years.

ii) Find the probability that at most 25% of the soil will lapse in any given 10 years.

7. A storekeeper wanted to buy a large quantity of light bults form two brands labelled A & B. He bought 100 bulbs from each bulbs brand & found by tasting that brand 'A' had mean life time 1120 hrs & s.d. 75 hrs & brand 'B' had mean life time 1062 hrs

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& s.d. 82 hrs. Find the 99% confidence limits for the difference in the average life of bulbs from two brands. (8)

8. A sample of 100 iron bars is said to be drawn from a large number of bars whose lengths are normally distributed with mean 4 ft & s.d. 0.6 ft. If the sample mean is 4.2 ft can the sample be regarded as a truly random sample. (7)

9. Sales of new electronic item is six stores before & after special promotional program are observed as follows: (8)

Sales before campaign	50	30	31	48	55	42
Sales after campaign	52	29	30	52	56	45

10. Strength tests carried out on samples of two yarns spurn to the same count gave the following results. (7)

Yarn 'A' Sample size 4 Sample mean: 52 Variance: 42
Yarn 'B' Sample size 9 Sample mean: 42 Variance: 56

Test significance of mean strengths of two sources which is preferable.

11. Random sample of weekly road accidents that occurred during the variances days of the week in Kathmandu valley is given below: (8)

Day:	Sun	Mon	Tue	Wed	Thu	Fri	Sat
No of accidents:	14	16	8	12	11	9	14

Test whether the accidents are uniformly distributed over the week.

12. The following table gives the age of blood pressure of 10 women. (8)

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Age (x):	56	42	36	47	49	42	60	72	63	55
Blood Pressure	147	125	118	128	145	140	155	160	149	150

(i) Find the correlation co-efficient between x & y.
(ii) Determine the regression equation of y on x.
(iii) Estimate the blood pressure of women whose age is 45 yrs.

13. Write short notes (9)

a) Method of primary data collection
b) Application of statistics in engineering.

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Madan Bhandari Memorial Academy Nepal
Internal Assessment

Level:Bachelor	**Semester : Fall**	**Year:2025**
Programme:BE	**Time: 3hrs**	**Full Marks:100**
Course: Probability and Statistics		**Pass Marks: 45**

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

1 a) The following data set represents the number of new computer accounts registered during ten consecutive days. 7

43,37,50,51,58,105,52,45,45,10

i. Compute the mean, median and standard deviation

ii. Draw a box and whisker plot and identify whether it is skewed or not.

b) After the implementation of an economic program to uplift the economic condition of a community following information were found 8

Monthly income (Rs.000)	4-6	6-8	8-10	10-12	12-14	14-16	16-18
After the plan(no. of families	8	65	37	15	15	5	5

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Construct an ogive to find

i. Find the number of families whose monthly income is between Rs.8000 to Rs. 14000

ii. Find the number of families whose monthly income is above Rs. 12000 25

a) A firm has 80% of its service calls made by a contractor and 10% of these calls result in customer complaints. The other 20% of the service call are made by their own employees, and these calls have a 5% complaint rate. Find the probability of a complaint. Also, using Bayes theorem to find the probability that a complaint was 7

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from a customer whose service was provided by the contractor.

b) Three bags contain 3 video cards and 2 network cards, 5 video cards and 6 network cards, 2 video cards and 4 network cards respectively. One card is drawn from each urn. Find the expected number of video cards. 8

3 a) Message arrive at an electronic message center at random times, with an average of 9 messages per hour. 5

i. What is the probability of receiving at least five messages during the next hour?

ii. What is the probability of receiving exactly five messages during the next hour?

b) The average diameter of a sample of 1000 pipes is 2.6 inches and standard deviation is 0.55. Assuming that the diameter of pipes is normally distributed. Find the number of pipes of diameter 5

i. Greater than 2 inches.

ii. Between 2 and 3.1 inches

c) A continuous random variable X has the following density function. 5

$$f(x) = kx^2 \quad for\ 0 < x < 1$$
$$= 0 \text{ otherwise}$$

i. Find the value of k

ii. $P(0<X<0.7)$

iii. $E(X)$

a) In a random sample of 300 households in a city 223 have computer. Set an approximate 95% and 99% limits to the true value of proportion of households with computer in the whole city. 7

b) A random sample of 200 bolts manufactured by machine A and of 100 bolts manufactured by machine B showed 19 and 5 defective bolts respectively. Test the hypothesis that machine B is performing better than A at 5% level of significance. 8

a) The sales data of steel for a steel factory in six shops before and after a special promotional campaign are as under: 7

b)

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Shops	A	B	C	D	E	F
Before campaign	53	28	31	48	50	42
After campaign	58	29	30	55	56	45

Can the campaign be judged to be a success? Test at 5% level of significance 8

c) Describe the importance of sampling. Two types of drugs were used on 5 and 7 patients for reducing their weight. Drug A was imported and drug B indigenous. The decrease in the weight after using the drugs for six months was as follows:

Drug A	12	13	11	14	10		
Drug B	9	12	14	15	10	9	8

Is there a significant difference in the efficiency of two drugs?

6 a) Scraps of iron were selected on the basis of their densities, X and their iron contents Y were measured. The results were as follows: 7

X	2.8	2.9	3.0	3.1	3.2	3.2	3.2	3.3	3.4
Y	27	23	30	28	30	32	34	33	30

i. Find the relationship between their densities ad their iron content

ii. Find the regression equation of y on x.

b) A Population consists of five members 2,3,6,8 and 11.

i. Draw all possible sample of size 2 that can be drawn from the population without replacement. 8

ii. Find mean and variance of population

iii. Find the mean of sampling distribution of means and show that it is equal to the population mean.

7 Write short notes on :(Any two) (10)

a) Sourve of data and its types.

b) Normal distribution and its characteristics

c) Criteria of good estimator.

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MADAN BHANDARI MEMORIAL ACADEMY NEPAL

Level: Bachelor
Programme: BE
Course: Probability and Statistics (Set B)

Year : 2080
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Following information shows the daily wage of workers of certain hydropower company , prepare suitable ogive that helps to give the answers of following questions 7

Daily Wages	0-20	20-40	40-60	60-80	80-100
No. of workers	41	51	64	38	7

i. About what wage above that 50% worker earn?

ii. What would be the daily wage limit of middle 30% workers?

b) The following information shows the income distribution. 8

Income (S 000)	0-10	10-20	20-30	30-40	40-50	50-60
No. of Person	5	10	18	23	7	6

Construct less than ogive. Also use it to find (i) the number of persons having income less than $35,000 and (ii) percentage of persons having income between $20,000 and $50,000.

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2. a) In an organization, there are 2 Civil engineers, 4 IT officers and 2 accountants. A committee of four members has to be formed. What is the probability that a committee contains i) one Civil engineer and 3 IT officers? ii) no Civil engineer? iii) at least one IT officers? 7

b) A Company has two plants to manufacture the screws. Out of 10000 screws, plant I manufactures 80% of the screws and plant II manufactures 20%. At plant I, 85% screws are rated standard quality. At plant II, only 65% screws are rated standard quality. 8

i. One screw is selected at random from the whole consignment and was found standard quality, What is the probability that the selected screw was manufactured by plant I?

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ii. What is the probability that the selected screw being defective?

3. a) The diameter of an electric cable say X , is assumed to be a continuous random variable with 7

$$f(x) = \begin{cases} 6x(1-x) & ;0 \le x \le 1 \\ 0 & ;otherwise \end{cases}$$

i. Check whether about function is p.d.f or not

ii. Find E(x) and v (x)

b) At a certain examination 10% of the student who appeared for the paper of statistics got less than 30 marks and 97% of the students got less than 80 marks. Assuning that the distribution of mark is Normal. Find 8

i. mean and variance of marks of student

ii. Find the probability tat randomly selected student get more than 70 marks

4. a) Messages arrive at an electronic message center at random 5

time, with an average of 9 messages per hour.

(i)What is the probability of receiving at least five messages

during the next hour?

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(ii) What is the probability of receiving exactly five messages

during the next hour?

b) The Joint Probability density function of two dimensional randon variable (x,y) is given by 5

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Shops	A	B	C	D	E	F
Before campaign	53	28	31	48	50	42
After campaign	58	29	30	55	56	45

Can the campaign be judged to be a success? Test at 5% level of significance 8

c) Describe the importance of sampling. Two types of drugs were used on 5 and 7 patients for reducing their weight. Drug A was imported and drug B indigenous. The decrease in the weight after using the drugs for six months was as follows:

Drug A	12	13	11	14	10		
Drug B	9	12	14	15	10	9	8

Is there a significant difference in the efficiency of two drugs?

6 a) Scraps of iron were selected on the basis of their densities, X and their iron contents Y were measured. The results were as follows: 7

X	2.8	2.9	3.0	3.1	3.2	3.2	3.2	3.3	3.4
Y	27	23	30	28	30	32	34	33	30

i. Find the relationship between their densities ad their iron content

ii. Find the regression equation of y on x.

b) A Population consists of five members 2,3,6,8 and 11.

i. Draw all possible sample of size 2 that can be drawn from the population without replacement. 8

ii. Find mean and variance of population

iii. Find the mean of sampling distribution of means and show that it is equal to the population mean.

7 Write short notes on :(Any two) (10)

a) Sourve of data and its types.

b) Normal distribution and its characteristics

c) Criteria of good estimator.

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2082/11/14			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V	1.5 hrs	

Subject: Probability and Statistics

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ ***Attempt All question.***
- ✓ *The figures in the margin indicate* ***Full Marks****.*
- ✓ *Assume suitable data if necessary.*

1. a) The polythene bags are taken randomly from two manufacturing company's A and B; and are tested by a prospective buyer for bursting pressure. The results are as follows. It is generally assumed that the bags having more uniform pressure will have long lifetime. If the prices are same, which manufacturing company bags would be preferred by the buyer and why? **7**

Bursting Pressure (in lbs.)	No. of Polythene bags	
	A	B
5 – 10	2	9
10 – 15	9	11
15 – 20	29	18
20 – 25	44	32
25 – 30	11	17
30 – 35	5	13

b) A potential buyer of light bulbs bought 50 bulbs of each of two brands upon testing these bulbs, he found that Brand A has a mean life of 1282 hours with a standard deviation of 80 hrs whereas B had a mean life of 1208 hrs with a standard deviation of 94 hrs can the buyers be quite certain that the two brands differ in quality? **8**

2. a) A population consists of 4 units A, B, C, and D with values 8, 2, 6, and 4.
   i. Construct a sampling distribution of sample mean by selecting samples of size 2 in random sampling without replacement. **8**
   ii. Find the mean and variance of the sampling distribution of the sample mean
   iii. Also examine whether the sample mean is an unbiased estimator of population mean.

OR

A random sample of 10 students had the following I. Q's: 70, 120, 110, 101, 83, 88, 95, 98, 107, and 100. Find the 95% confidence interval for the mean I.Q values lie.

b) A random sample of 500 apples was taken from a large consignment and 60 were found to be bad. Obtain the 95% and 99% confidence limits for the percentage number of good apples in the consignment. **7**

3. a) An experiment was run to determine the effectiveness of a new type of drug on blood pressure. Ten persons have their blood pressure measured before and after the drug is given. The result of the experiment are as follows: **7**

Before	116	118	120	124	128	130	131	134	136	137
After	119	124	126	128	121	135	137	138	139	135

Test the hypothesis that new drug raises the blood pressure at $\alpha = 0.05$.

b) Given the following data of CPU time required (Y) and the number of disk I/O operations (X). **8**

Time in sec. (Y)	24	38	42	50	60	30	20	25	40	39
Number (X)	398	390	410	502	590	305	210	252	398	392

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i. Compute the coefficient of correlation between X and Y.
ii. Estimate the CPU time requirement for 550 disk I/O operations.
iii. Find coefficient of determination and interpretation.

Write short notes on any ONE: $1 \times 5 = 5$

a) Criteria of good estimator.
b) Errors in hypothesis testing.

Best Wishes!!!

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor | Semester: V | Year: 2025
Programme : BE Computer | Time: 3 hours | Full Marks: 100
Course: Probability & Statistics(old) | Pass Marks: 45

Pre-Board Examination - 2082 (Fall 2025)

Candidates are requested to give their answers in their own words as far as practicable. Figure in the margin indicates full marks.

Attempt all the questions:

1. a)Over a period of 40 days the percentage relative humidity in a vegetable storage building was measured. Mean daily values were recorded as shown below:

60	63	64	71	67	73	79	80	83	81
86	90	96	98	98	99	89	80	77	78
71	79	74	84	85	82	90	78	79	79
78	80	82	83	86	81	80	76	66	74

i)Prepare a steam and leaf display for these data. (8)

ii)Draw a box plot for these data .

b) An analysis of monthly wages paid to the workers in two firma A and B belonging to the same industry gives the following results. (7)

	Firm A	Firm B
No. of workers	500	600
Average monthly wages(Rs)	186	175
Variance of distribution of wages(Rs)	81	100

i) Which Firm A or B has a larger wage bill?

ii) in which firm A or B is there grater variability in industrial wages?

2. A husband and wife appear in an interview for two vacancies in the same post. The probability of husband's selection is 1/7 and that of wife's selection is 1/5 . What is the probability that

i) both of them will be selected.

ii) only one of them will be selected.

iii) none of them will be selected.

iv) husband will be selected but not wife.

v) wife will be selected but not husband? (7)

b)state and prove Baye's theorem for conditional probability. (8)

3. a)A random variable X has the following probability function.

X	2	3	4	5	6	7
F(x)	0.1	k	0.2	2k	0.3	k

Find the value of K and calculate mean and variance. (7)

b)An office switchboard receives telephone calls at the rate of 3 calls per minute on an average. If receiving of calls follows a Poisson distribution, find the probability of receiving : (8)

i) no calls in one minute interval.

ii) at least 3 call in a one minute interval.

iii) at most 2 calls in 5 minute interval.

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a)The average height of students of universal engineering and science college is 165 cm and standard deviation is 10 cm. find the percentage of students whose height is i)less then 172 cm.ii)between 150 and 180 cm.iii)more then 174 cm.iv)for what value of variable X,$P(X<x_1)=0.10$ holds ? (8)

b)The joint probability function of random variable X and Y is given by

$$f(x,y)=\begin{cases}4xy & for\ 0<x<1, 0<y<1\\ 0 & otherwise\end{cases}$$

i) verify the joint probability function is pdf.
ii) find p(X<0.5 ,Y<0.8). iii)Find marginal density of X and Y.
iii) Are X and Y independent? (7)

5. The quality control Engineer at a light bulb factory needs to estimate the average life of a large shipment of light bulbs . The process standard deviation is known to be 100 hours . A random sample of 64 lights bulbs indicated a sample average life of 350 hours.
   i) Calculate the standard error of mean .
   ii) Set up a 95% confidence limits for the true average life of light bulbs in this shipment. (8)

   b) A study shows that 16 of 200 tractors produce on one assembly line required extensive adjustment before likely they could be shipped , while same was true for 14 of 400 tractors produced on another assembly line . At the 0.05 level of significance , does this support claim that the second production line does superior work?
   (7)

6. a) an I.Q. test was administered to 5 persons before and after they were trained . The result are given below:

candidates	1	2	3	4	5
IQ before Training	110	120	123	132	125
IQ after training	120	118	125	136	121

is the training effective at 5% level of significance?(7)

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b)The following table gives the age X and blood pressure Y of 10 women .

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	149	150

i) Find correlation of coefficient between age pressure .
ii) Fit a regression equation of X on Y. (8)

7. Write short notes on (Any two) [2*5=10]

a) Probability mass function and probability density function.
b) Central limit theorem.
c) Error in hypothesis testing.

THE END

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UNITED TECHNICAL COLLEGE
Semester-Fall

Level: Bachelor
Program: B.E. Comp
Course: Probability and Statistics

Year: 2025
Full Marks: 50
Time: 1.5hours

Candidates are required to give their answers in their own words as far as practicable. The figure in the margin indicates full marks.

Answer ALL the questions.

1 a The lives of two models (A and B) of refrigerators in a recent survey are shown below: 7

Life (No of years)		0-2	2-4	4-6	6-8	8-10	10-12
No of refrigerators	Model A	5	16	13	7	5	4
	Model B	2	7	12	19	9	1

Which model has greater uniformity?

b The contents of Urns I, II and III are as follows: 8

1 white, 2 black and 3 red balls
2 white, 1 black and 1 red ball
4 white, 5 black and 3 red balls

One urn is chosen at random and two balls are drawn. they happen to be white and red. what is the probability that they come form Urn I , II and III?

2 a A random variable X is distributed at random between the values 0 and 1 so that its pdf is 5

$$f(x) = \begin{cases} kx^3(4-x)^2 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

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Where, k is a constant. Find the value of k, using this value of k find mean and variance of distribution.

b A car hire firm has 2 cars that it hires out day by day. The number of demands for a car on each day is distributed as Poisson distribution with mean 1.5. Calculate the proportion of days on which i) Neither car is used ii) some demand is refused 5

c A recent study of how Nepalese spend their leisure time surveyed workers employed more than 5 years. They determined the probability an employee has 2 weeks of vacation time to be 0.45, one week of vacation time to be 0.1 and three or more weeks to be 5

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0.2. Suppose 12 workers are s... ...random. Answer the following questions.

i. What is the probability that at most two of the workers have three or more weeks of vacation time?
ii. What is the probability that at least 3 workers have one week of vacation time?

3 a The income of a group of 20,000 persons was found to be normally distributed with mean Rs. 7,500 and standard deviation Rs. 500. Find 7

i. The lowest income of richest 20% of persons
ii. The highest income of poorest 10% of persons.

b Two random variables X and Y have the following probability density function. 8

$$f(x,y) = \begin{cases} k(4-x-y) & 0<x<2, 0<y<2 \\ 0 & \text{otherwise} \end{cases}$$

otherwise

Find

i. The constant k
ii. Marginal density function of X and Y
iii. Conditional probability density function of X given Y
iv. E(X) and V(X)

4 a From a population of 540, a sample of 60 individuals is taken. From this sample, the mean is found to be 6.2 and standard deviation 1.368. 7

i. Find the estimated standard error of the mean
ii. Construct 96% confidence interval for the mean

b The average weekly wages of a sample of 200 workers in industry A was Rs. 1150 with standard deviation of Rs. 100. The weekly wages of a sample of 300 workers in industry B was Rs. 1,000 with standard deviation of Rs. 50. Can we consider the weekly wages paid by industry A is higher than those paid by industry B? 8

5 a In a random sample of 1000 houses in a certain city it was found that 318 own color T.V. In the neighboring city it was found that 450 out of 1200 houses own color T.V. Test whether there is 7

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significant difference between the two samples of housing with respect to using color T.V. at 5% level of significance

b The average gas mileage for 12 cars of brand A was 27.2 miles per gallon and standard deviation was 3.8 miles per gallon. The 9 brand B cars that were tested averaged 32.1 miles per gallon and the standard deviation was 4.3 miles per gallon. At 0.01 level of significance should it conclude that brand A cars have lower average gas mileage than do brand B cars? 8

6 a Where people turn to for news is different for various age groups. A study indicated where different age group primarily get their news. 7

Media	Age group		
	Under 36	36-50	Above 50
Local TV	17	11	13
National TV	7	10	12
Radio	7	9	10
Local Newspaper	5	7	10
Internet	9	8	7

At the 0.05 level of significance, is there evidence of a significant relationship between the age group and where people primarily get their news?

b Shear stress (y) and rate of shear (x) can be measured for a liquid in a viscometer. For 12 pairs of values the data can be summarized as 8

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$\sum X=132, \quad \sum Y=151.7, \quad \sum X^2=1944, \quad \sum Y^2=2570.48, \quad \sum XY=2233.2$

i. Find the linear regression equation of y on x
i. What is the correlation coefficient?
i. What fraction of the variance of y is explained by x?

7 Write short notes on any One. 5×2

a Errors in Hypothesis Testing
b Describe Types of Sampling
c Importance of Statistics in Engineering

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POKHARA UNIVERSITY

Level: Bachelor | Final Internal Examination | Year: 2026

Programme: BE | Full Marks: 100 Pass Marks: 45

Course: Probability and Statistics | Time: 3 hrs

Candidates are required to give their answers in their own words as far as practicable. The figures in margin indicate full marks.

Attempt all the questions.

1. a) An incomplete distribution of number of students according to marks secured by them is given below. The median and mode of the distribution of marks are 25 and 24 respectively. Calculate the missing frequencies and also find seventh deciles of the series graphically. 7

Marks:	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50
No. of students:	14	?	27	?	15

b) The chances that A, B and C becoming a chief program developer of a certain software development company are 4:2:3. The probability that new software will be developed if A, B and C become chief developer of that company are 0.3, 0.5 and 0.8. If the new software has been developed, then what is the probability that Mr. C is appointed as chief developer of the company? 8

2. a) The number of hardware failure of a computer system in a week of operation has the following probability mass function

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No. of failures:	0	1	2	3	4	5	6
Probability:	0.18	0.28	0.25	0.16	0.04	0.06	0.03

Find the standard deviation of the number of failures in a week. 7

b) The phone lines to an airline reservation system are occupied 40% of the time. Assume that the events the lines are occupied on successive calls are independent. Assume that 10 calls are placed to the airline then what is the probability that i) Exactly 3 calls the lines are occupied? ii) For at least one call the call lines are not occupied? 8

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3.a) What is standard normal variate? From past experience it is known that time to download on certain university webpage is normally distributed with mean of 9 seconds with standard deviation of 2 seconds. What is the probability that the download will be completed i) under 3.5 seconds ii) more than 3 seconds? 8

Or,

A set of final examination grades in an introductory statistics course was found to be normally distributed with mean of 73 and standard deviation of 8. i) What is the probability of getting a grade no higher than 91 on this exam? ii) Only 5% of the students taking the test scored than what grade? 8

b) Out of 3000 families 4 children each, assume boys and girls are equally probable. How many families would you expect to have i) at most 2 girls ii) at least one boy? 7

4.a) The joint probability distribution of X and Y are:

$f(x, y) = 2 - x - y, \quad 0 < x < 1, \quad 0 < y < 1$

$= 0$, otherwise

Find marginal probability density function of x and y, are X and Y independent? 7

b) A population consists of the values 7, 6, 8, 4 and 10. Prove that sample mean of size 2 is unbiased estimate of population mean. 8

5.a) The following are the Brinell hardness values obtained for eight samples of magnesium alloys:

Hardness of alloys: 107, 148, 123, 165, 102, 119, 115, 125

Find 95% confidence interval of the mean Brinell hardness of the magnesium alloys. 8

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b) In random sample 1000 houses in a certain city it was found that 318 have internet facilities. In the neighboring city it was found that 450 houses out of 1200 houses have internet facility. Test whether there is significance difference between houses of two cities having internet facilities at 5% level of significance? 7

Or,

b) A quality control engineer at a light bulb factory needs to estimate the average life of a large shipment of light bulbs. The process standard deviation is known to be 100 hours. A random sample 64 light bulbs indicated a sample mean of 350 hours. Would you claim

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that the manufacture has the right to state that the light bulbs last an average of 400 hours at 5% level of significance? 7

6.a) The score of 8 candidates priori and after training are given below.

Priori:	82	48	36	54	72	84	90	65
After:	92	42	45	81	84	75	86	70

Was training effective at 5% level of significance? 7

b) The following data presents the marks obtained by students in mathematics and physics.

Marks in mathematics:	80	60	72	40	82	90
Marks in physics:	90	72	75	65	50	80

Fit the regression equation of marks of physics on marks of mathematics and predict the marks of mathematics whose marks of physics are 52. 8

7. Write short note on any two: 2*5 = 10

a) Errors in hypothesis testing

b) Interval estimation

c) Karl Pearson's correlation coefficient

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Gandaki College of Engineering and Science

Level: Bachelors | Semester: Fall | Year: 2025
Program: BE Computer/Software | **FINAL ASSESSMENT** | Full Marks: 100
Course: Probability and Statistics | Pass Marks: 45

Candidates are required to give their answers in their own words as far as practicable.

Attempt all the questions.

1. a) The life of two models of Ceiling Fans in a recent survey is given below. (8)

Life (in Years)	Number of Ceiling Fan	
	Model 1	Model 2
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	9
10-12	4	1

i. What is the average life of each model of Ceiling Fans?
ii. Which model of Ceiling fan has greater uniformity?

b) In a class of 75 students, 15 students were considered to be very intelligent, 45 are medium and rest are below average. The probability that a very intelligent student fails in an examination is 0.005, the medium is 0.05 and the below average is 0.15. If a student passed in an examination, what is the probability that he/she is below average? (7)

2. a) In a certain city a survey of 600 families about the fuel for cooking revealed that 230 use LPG, 175 use Induction and 40 use both. Then find the probability that a family selected uses: (8)
i. At least one of the fuel
ii. Neither of the fuel
iii. Exactly one type of fuel

b) A coin is tossed until a head appears. What is the expectation number of tosses required? (7

OR

A random variable X has the following probability function:

X	2	3	4	5	6	7
P(X=x)	0.1	K	0.2	2k	0.3	k

i. Find the value of k
ii. Find E(X) and V(X)
iii. Find the distribution function of X

3. a) An aircraft has 116 seats. From long experience, on average 2.5% of the people who have bought tickets do not arrive for the flight. The airlines sell 120 tickets for a flight to Pokhara. Calculate the probability that: (8
i. More than 118 people arrive for the flight?
ii. There are empty seats on the flight?

b) In an examination 15% of the students got first class marks above 60 and while 40 % failed marks below 40. Assuming the marks to be normally distributed, estimate the mean and standard deviation. Also, find its coefficient of variation. (

4. a) Suppose that X is a random variable whose probability density function is given by: (

$$f(x, y) = C(4x - 2y^2), 0 < x < 2, 0 < y < 1$$

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i. Determine the value of C
ii. Determine the cumulative probability of X
iii. Find the marginal density function of X and Y
iv. Find the conditional density function of X given Y=y

b) Define Estimation. A hotel manager in Kathmandu wants to know the hotel's average daily registration. The following table presents the number of guests registered for 14 randomly selected days: (8)

61, 57, 60, 57, 54, 58, 55, 52, 65, 40, 64, 70, 22, 56.

Calculate the sample mean, standard error, and 95% confidence interval of the population mean.

OR

Define point and interval estimation. A certain Cinema Hall sampled 55 people who viewed Hero Number One and asked them whether they planned to see it again. Only 10 of them believed the film was worthy of a second look.

i. Estimate the standard error of the proportion of movie viewers who will view the film a second time.
ii. Construct a 95% confidence interval for this proportion.

5. a) A population consists of the values 1, 2, 3, 6, 8. Prove that the sample mean of size 3 is an unbiased estimate of the population mean (without replacement). (7)

OR

Two independent random samples of 70 boys and 55 girls from a normal population were drawn, and their mean weights were 62 kg and 46 kg, with variances of 16 and 12, respectively. Construct a 95% confidence interval for the difference between the mean weight of boys and girls.

b) A random sample of 400 males and 600 females was asked whether they would like to have an airport near their residence. 200 men and 325 women were in favor of the proposal. Test the hypothesis that the proportion of men and women is in favor of the proposal at 5% level of significance. (8)

6) a) A principal of the college thought that memory power training is better for the learning process of students, so memory power training was launched. The memory capacity of the students was noted before and after training. The data is recorded as: (7)

Before Training	56	42	36	47	49	42	60	72	63	
After Training		147	125	118	128	145	140	155	160	149

Is the training effective at 5% level of significance?

b) The following table gives the age Xand blood pressure Y of 10 women. (8)

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	140	150

i. Find the correlation coefficient between age and blood pressure.
ii. Fit a linear regression equation of X on Y.

7 Write short notes on: (any two) (10)

i. Bayes Rule
ii. Binomial Distribution
iii. Type I and Type II Errors in Hypothesis testing.

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Pokhara University
Everest Engineering College
Final Assessment
Semester - Fall

Level: Bachelor
Program: BE CMP (5th Semester)
Faculty: Science and Technology
Subject: Probability & Statistics (New)

Year: 2025
F.M: 100
P.M: 45
Time: 3 hrs.

Attempt all the questions.

1. (a) The factories produces two type of car batteries i.e. battery A and battery B. An experiment shows the life of batteries in days which were recorded as follows.

Life in days	500-700	700-900	900-1100	1100-1300	1300-1500
Battery A	5	11	26	10	8
Battery B	4	30	12	8	6

Compare the variability of two type of batteries using coefficient of variation. [7]

(b) Stare Baye's Theorem. A first step towards identifying spam is to create a list of words that are more likely to appear in spam than in normal messages. suppose a specified list of words is available and that your data base of 5000 messages contains 1700 that are spam. Among the spam messages, 1343 contain words in the list. Of the 3300 normal messages, only 297 contain words in the list. Obtain the probability that a message is spam given that the message contains words in the list. [8]

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2. (a) The probability density function of continuous random variable X is given by $f(x) = Kxe^{-4x^2}$. $x \geq 0$. Then find the value of K. Also, find F(120), E(X) and V(X).[8]

(b) In Binomial distribution consisting of 5 independent trials. the probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

(OR)

In a certain factory turning out razor blades, there is a small chance 1/500 for any blade to be defective. The blades are supplied in a packet of 10. Use Poisson distribution to calculate the approximate number of packets containing on defective, one defective, two defectives and 3 defectives blades in a consignment of 10,000 packets. [7]

3. (a) Define Gamma distribution and find its mean and variance. [7]

(b) In an examination, 10% of the candidates obtained marks below 20 and 5% of candidates obtained over 75 marks. Assuming that the marks are normally distributed, find the mean and standard deviation of the distribution? Also, find the probability that a selected candidate will obtain the marks more than 90 ? [8]

4. (a) Suppose that the random variables X and Y have the joint

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pdf

$$f(x,y) = \begin{cases} k(2x+y) & \text{if } 0 < x < 2,\ 0 < y < 3 \\ 0 & \text{if otherwise} \end{cases}$$

i. Evaluate the constant k.

ii. find the marginal probability density function of X and Y.

iii. Find the conditional probability distribution of Y given $X = x$.

iv. Find $P(X + Y < 3)$ and $P(X < 1/Y < 3)$

v. Are X and Y independent? [8]

(b) A random sample of 500 bulbs was taken from the manufacturing company among them, 65 were found to be defective. Construct 99% confidence interval for the population proportion of defectiveness. Also, estimate the standard error of the proportion. [7]

5. (a) What are the characteristics of good estimator? A random sample of 8 envelopes is taken from letter box of a post office and their weights in grams are found to be 12.1, 11.9, 12.3, 11.5, 11.9, 12.1, 12.4, 12.1. Find 99 % fiducial limits (confidence interval) for the mean weigh of envelope received at that post office. [7]

(b) What is critical value?. In a random sample of 1000 houses in a certain city it was found that 318 have internet facilities. In the neighbouring city it was found that 450 houses out of 1200 houses have internet facility. Test whether there is significance difference between houses of two cities having internet facilities at 5% level of significance? [8]

6. (a) An experiment was run to determine the effect of a new type drug on blood pressure. 10 persons have their blood pressure measured before and after the drug is given and the result of the experiment are as follows:.

Persons	1	2	3	4	5	6	7	8	9	10
Before	110	118	120	124	128	130	131	134	136	137
After	110	124	136	128	121	135	137	138	139	135

Test the hypothesis that new drug raise blood pressure at $\alpha = 5\%$ level of significance? [7]

(OR)

In an experiment to study the dependence of hypertension and smoking habits, the following data were taken on 180 individuals.

Cases	Non smokers	Moderate smokers	Heavy smokers
Hypertension	21	36	30
No Hypertension	48	36	10

Test the hypothesis that presence or absence of hypertension is independent of smoking habits at 5% level of significance?

(b) The following table presents shear strengths(in kN/mm) and weld diameters (in mm) for a sample of spot welds.

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Diameter	4.2	4.4	4.6	4.8	5.0	5.2	5.4
Strength	51	54	69	81	75	79	89

i. Find correlation coefficient and then test whether it is significant or not. Also, compute the coefficient of determination and interpret the result.

ii. Compute the least square regression line for predicting strength from diameter and then predict the strength for a diameter of 5.5 mm. Also, compute standard error of estimate. [8]

7. Write short notes on :**(Any Two)**

(a) Probability and Non probability sampling

(b) Types of error in hypothesis testing

(c) Application of statistics in engineering [2 × 5 = 10]

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